FIT3080 Artificial Intelligence
Solutions to Practice Questions in MDPs and Reinforcement Learning

1. **[MDP and RL]** Consider a simple MDP with two states: S_1 and S_2, two actions: A and B, a discount factor γ of $1/2$, a reward function R given by:

$$R(s,a,s') = \begin{cases} 2 & \text{if } s = S_1, a = A, s' \in \{S_1, S_2\} \\ 3 & \text{if } s = S_1, a = B, s' \in \{S_1, S_2\} \\ -2 & \text{if } s = S_2, a = A, s' \in \{S_1, S_2\} \\ -3 & \text{if } s = S_2, a = B, s' \in \{S_1, S_2\} \end{cases}$$

The transition function is specified by the following table:

s	a	s'	$\Pr(s' \mid s, a)$
S_1	A	S_1	$4/5$
S_1	A	S_2	$1/5$
S_1	B	S_1	$1/2$
S_1	B	S_2	$1/2$
S_2	A	S_1	$1/2$
S_2	A	S_2	$1/2$
S_2	B	S_1	$2/3$
S_2	B	S_2	$1/3$

The Bellman equations are as follows:

$$V^*(s) = \max_a Q^*(s,a)$$
$$Q^*(s,a) = \sum_{s'} T(s,a,s') \Big[R(s,a,s') + \gamma V^*(s') \Big]$$
$$V^*(s) = \max_a \sum_{s'} T(s,a,s') \Big[R(s,a,s') + \gamma V^*(s') \Big]$$

(a) Suppose that after i iterations of Value Iteration, the values are now as shown in the following table. Perform a single round of value iteration, filling in the resulting state values, on the basis of the provided initial values V_i (rather than starting from zero state values).

SOLUTION:

s	V_i	V_{i+1}
S_1	1	3
S_2	-1	-2

$$\begin{aligned} V_{i+1}(S_1) &= \max_a Q_{i+1}(S_1, a) \\ V_{i+1}(S_1) &= \max\{Q_{i+1}(S_1, A), Q_{i+1}(S_1, B)\} = \max\{2.3, 3\} = 3 \\ V_{i+1}(S_2) &= \max_a Q_{i+1}(S_2, a) \\ V_{i+1}(S_2) &= \max\{Q_{i+1}(S_2, A), Q_{i+1}(S_2, B)\} = \max\{-2, -2.833\} = -2 \end{aligned}$$

$$
\begin{aligned}
Q_{i+1}(S_1, A) &= T(S_1, A, S_1)[R(S_1, A, S_1) + \gamma V(S_1)] + T(S_1, A, S_2)[R(S_1, A, S_2) + \gamma V(S_2)] \\
Q_{i+1}(S_1, B) &= T(S_1, B, S_1)[R(S_1, B, S_1) + \gamma V(S_1)] + T(S_1, B, S_2)[R(S_1, B, S_2) + \gamma V(S_2)] \\
Q_{i+1}(S_2, A) &= T(S_2, A, S_1)[R(S_2, A, S_1) + \gamma V(S_1)] + T(S_2, A, S_2)[R(S_2, A, S_2) + \gamma V(S_2)] \\
Q_{i+1}(S_2, B) &= T(S_2, B, S_1)[R(S_2, B, S_1) + \gamma V(S_1)] + T(S_2, B, S_2)[R(S_2, B, S_2) + \gamma V(S_2)]
\end{aligned}
$$

$$
\begin{aligned}
Q_{i+1}(S_1, A) &= 4/5[2 + 1/2 \times 1] + 1/5[2 + 1/2 \times (-1)] = 2.3 \\
Q_{i+1}(S_1, B) &= 1/2[3 + 1/2 \times 1] + 1/2[3 + 1/2 \times (-1)] = 3 \\
Q_{i+1}(S_2, A) &= 1/2[-2 + 1/2 \times 1] + 1/2[-2 + 1/2 \times (-1)] = -2 \\
Q_{i+1}(S_2, B) &= 2/3[-3 + 1/2 \times 1] + 1/3[-3 + 1/2 \times (-1)] = -2.833
\end{aligned}
$$

(b) Perform a single round of Q-value iteration, filling-in the resulting Q-values in the following table. Use the specified initial value function Q_i, rather than starting from all zero Q-values. Only compute the entries not labeled with "skip".

SOLUTION:

s	a	Q_i	Q_{i+1}
S_1	A	2	2.6
S_1	B	1	3
S_2	A	-3	skip
S_2	B	-2	skip

$$Q_{i+1}(s, a) = \sum_{s'} T(s, a, s')[R(s, a, s') + \gamma max_{a'} Q_i(s', a')]$$

$$
\begin{aligned}
Q_{i+1}(S_1, A) &= T(S_1, A, S_1)[R(S_1, A, S_1) + \gamma \max_{a'} Q_i(S_1, a')] + \\
&\quad T(S_1, A, S_2)[R(S_1, A, S_2) + \gamma \max_{a'} Q_i(S_2, a')] \\
Q_{i+1}(S_1, B) &= T(S_1, B, S_1)[R(S_1, B, S_1) + \gamma \max_{a'} Q_i(S_1, a')] + \\
&\quad T(S_1, B, S_2)[R(S_1, B, S_2) + \gamma \max_{a'} Q_i(S_2, a')]
\end{aligned}
$$

$$
\begin{aligned}
Q_{i+1}(S_1, A) &= 4/5[2 + 1/2 \times \max\{2, 1\}] + 1/5[2 + 1/2 \times \max\{-3, -2\}] = 2.6 \\
Q_{i+1}(S_1, B) &= 1/2[3 + 1/2 \times \max\{2, 1\}] + 1/2[3 + 1/2 \times \max\{-3, -2\}] = 3
\end{aligned}
$$

(c) Suppose that we did not know the transition function T. We still need to find an optimal policy, so we make use of Q-learning, with a learning rate $\alpha = 1/4$ and a discount factor of $\gamma = 1/2$.

We limit our learning episodes to at most 5 steps, and our first episode is:

$$[S_1, A, S_1, A, S_2, B, S_2, A, S_1, B, S_1]$$

Initializing the $Q(s, a)$ to zero, and using the rewards from item (a), what would be the new values for $Q(s, a)$ after observing this episode?

The Q-learning equation is:

$$Q(s_t, a_t) \leftarrow (1 - \alpha) \cdot Q(s_t, a_t) + \alpha \cdot \left(r_t + \gamma \cdot \max_a Q(s_{t+1}, a)\right)$$

SOLUTION:

t	$\langle s, a, s', r\rangle$	$Q(S_1, A)$	$Q(S_1, B)$	$Q(S_2, A)$	$Q(S_2, B)$
0	$\langle -, -, S_1, 0\rangle$	0	0	0	0
1	$\langle S_1, A, S_1, 2\rangle$	1/2	0	0	0
2	$\langle S_1, A, S_2, 2\rangle$	7/8	0	0	0
3	$\langle S_2, B, S_2, -3\rangle$	7/8	0	0	-3/4
4	$\langle S_2, A, S_1, -2\rangle$	7/8	0	-25/64	-3/4
5	$\langle S_1, B, S_1, 3\rangle$	7/8	55/64	-25/64	-3/4

t $\langle s, a, s', r\rangle$	$sample = r + \gamma \max_{a'} Q(s', a')$	$Q(s, a) \leftarrow (1 - \alpha) \cdot Q(s, a) + \alpha \cdot sample$
1 $\langle S_1, A, S_1, 2\rangle$	$sample = 2 + 1/2 \times 0 = 2$	$Q(S_1, A) \leftarrow 3/4 \times 0 + 1/4 \times 2 = 1/2$
2 $\langle S_1, A, S_2, 2\rangle$	$sample = 2 + 1/2 \times 0 = 2$	$Q(S_1, A) \leftarrow 3/4 \times 1/2 + 1/4 \times 2 = 7/8$
3 $\langle S_2, B, S_2, -3\rangle$	$sample = -3 + 1/2 \times 0 = -3$	$Q(S_2, B) \leftarrow 3/4 \times 0 + 1/4 \times (-3) = -3/4$
4 $\langle S_2, A, S_1, -2\rangle$	$sample = -2 + 1/2 \times 7/8 = -25/16$	$Q(S_2, A) \leftarrow 3/4 \times 0 + 1/4 \times (-25/16) = -25/64$
5 $\langle S_1, B, S_1, 3\rangle$	$sample = 3 + 1/2 \times 7/8 = 55/16$	$Q(S_1, B) \leftarrow 3/4 \times 0 + 1/4 \times (55/16) = 55/64$

2. **[MDP]** To decide whether to save or invest in stocks, we are given the following information. If you have \$1000+,

   - 1 year of saving has a 50% chance of gaining \$1000, and a 50% chance of maintaining your current amount.
   - 1 year of investing has a 75% chance of gaining \$2000, and a 25% chance of losing \$1000.

   You currently have \$1000, and a goal to save at least \$4000. Achieving your goal has a reward of 8, while losing all your money (\$0) has a reward of -16. In both cases you have to stop investing. In addition, reaching every other state has a reward of -1. You do not use discounting ($\gamma = 1$).

   (a) Express the problem stated above as a Markov Decision Process. What are the States, Actions, Transitions and Rewards?

   SOLUTION:
   States = {\$0, \$1000, \$2000, \$3000, \$4000}
   Actions = {save, invest}
   Transitions

State	**Action**	**Next State**	**Probability**
s	save	$s + 1000$	0.5
s	save	s	0.5
s	invest	$s + 2000$	0.75
s	invest	$s - 1000$	0.25

   Rewards

$\langle s, a, \$4000\rangle$	8
$\langle s, a, \$0\rangle$	-16
$\langle s, a, \$0 < s' < \$4000\rangle$	-1

(b) What is the start state? What are the terminal states?

SOLUTION:

Start state: $s = \$1000$

Terminal states:

$s = \$4000$ (objective reached), $V^*(\$4000) = 0$

$s = \$0$ (ruin), $V^*(0) = 0$

(c) The Bellman equations are as follows:

$$V^*(s) = \max_a Q^*(s,a)$$

$$Q^*(s,a) = \sum_{s'} T(s,a,s')\Big[R(s,a,s') + \gamma V^*(s')\Big]$$

$$V^*(s) = \max_a \sum_{s'} T(s,a,s')\Big[R(s,a,s') + \gamma V^*(s')\Big]$$

Write down the Bellman equation for the optimal value $V^*(s)$ of the starting state in terms of the values $V^*(s')$ of the states reachable from the initial state, simplifying as much as possible.

SOLUTION:

$V^*(\$1000) = \max\{Q(\$1000, save), Q(\$1000, invest)\}$

$Q(\$1000, save) =$

$= T(\$1000, save, \$2000)[-1 + V^*(\$2000)] + T(\$1000, save, \$1000)[-1 + V^*(\$1000)]$

$= 0.5[-1 + V^*(\$2000)] + 0.5[-1 + V^*(\$1000)]$

$= -1 + 0.5V^*(\$2000) + 0.5V^*(\$1000)$

$Q(\$1000, invest) =$

$= T(\$1000, invest, \$3000)[-1 + V^*(\$3000)] + T(\$1000, invest, \$0)[-16 + V^*(\$0)]$

$= 0.75[-1 + V^*(\$3000)] + 0.25[-16 + V^*(\$0)]$

$= -4.75 + 0.75V^*(\$3000) + 0.25V^*(\$0) = -4.75 + 0.75V^*(\$3000)$

(d) The formula for the policy improvement step is

$$\pi_{k+1}(s) = \arg\max_a \sum_{s'} T(s,a,s')[R(s,a,s') + \gamma V^{\pi_k}(s')]$$

Given policy π with the following values $V^\pi(\$1000) = -2$, $V^\pi(\$2000) = 0$ and $V^\pi(\$3000) = 4.5$, what would be the policy $\pi'(s)$ for the start state \$1000 after a policy improvement step?

SOLUTION:

$Q^{\pi_{k+1}}(\$1000, save) =$

$= T(\$1000, save, \$2000)[R(\$1000, save, \$2000) + V^{\pi_k}(\$2000)] +$
$T(\$1000, save, \$1000)[R(\$1000, save, \$1000) + V^{\pi_k}(\$1000)]$

$= 0.5[-1 + 0] + 0.5[-1 - 2] = -2$

$$
\begin{aligned}
&Q^{\pi_{k+1}}(\$1000, invest) = \\
&\quad = T(\$1000, invest, \$3000)[R(\$1000, invest, \$3000) + V^{\pi_k}(\$3000)] + \\
&\quad\quad T(\$1000, invest, \$0)[R(\$1000, invest, \$0) + V^{\pi_k}(\$0)] \\
&\quad = 0.75[-1 + 4.5] + 0.25[-16 + 0] = -1.375
\end{aligned}
$$

So, $\pi'(\$1000) = invest$.

3. [**MDP and RL**] Consider an autonomous robot which can either move fast or slow in any time step. Moving fast generally gives a reward of +2, while moving slow gives a reward of only +1. However, the robot must also take into account its internal temperature, which can be either hot or ok . Driving slow tends to lower the temperature, while driving fast tends to raise it. If the robot is hot, there is a danger if it overheating, at which point it must stop, cool down, and make repairs. The MDP transitions and rewards are specified as follows:

s	a	s'	$T(s,a,s')$	$R(s,a,s')$
ok	slow	ok	1	+1
ok	fast	ok	0.5	+2
ok	fast	hot	0.5	+2
hot	slow	ok	1	+1
hot	fast	hot	0.5	+2
hot	fast	ok	0.5	-10

The Bellman equations appear below:

$$
\begin{aligned}
V^*(s) &= \max_a Q^*(s,a) \\
Q^*(s,a) &= \sum_{s'} T(s,a,s')\Big[R(s,a,s') + \gamma V^*(s')\Big] \\
V^*(s) &= \max_a \sum_{s'} T(s,a,s')\Big[R(s,a,s') + \gamma V^*(s')\Big]
\end{aligned}
$$

(a) Run two rounds of value iteration in the table below, using a discount of 0.8. You may skip the entry labeled "skip".
SOLUTION:

s	V_0	V_1	V_2
ok	0	2	3.2
hot	0	1	skip

$$V_{i+1}(s) \leftarrow \max_a \sum_{s'} T(s,a,s')[R(s,a,s') + \gamma V_i(s')]$$

Iteration 1:

$$
\begin{aligned}
V_1(\text{ok}) &\leftarrow \max\{Q(ok, fast), Q(ok, slow)\} = \max\{2, 1\} = 2 \\
V_1(\text{hot}) &\leftarrow \max\{Q(hot, fast), Q(hot, slow)\} = \max\{-4, 1\} = 1
\end{aligned}
$$

$$
\begin{aligned}
Q(ok, fast) &\leftarrow T(ok, fast, ok)[R(ok, fast, ok) + \gamma V_0(ok)] + \\
&\quad T(ok, fast, hot)[R(ok, fast, hot) + \gamma V_0(hot)] \\
Q(ok, slow) &\leftarrow T(ok, slow, ok)[R(ok, slow, ok) + \gamma V_0(ok)] + \\
&\quad T(ok, slow, hot)[R(ok, slow, hot) + \gamma V_0(hot)] \\
Q(hot, fast) &\leftarrow T(hot, fast, ok)[R(hot, fast, ok) + \gamma V_0(ok)] + \\
&\quad T(hot, fast, hot)[R(hot, fast, hot) + \gamma V_0(hot)] \\
Q(hot, slow) &\leftarrow T(hot, slow, ok)[R(hot, slow, ok) + \gamma V_0(ok)] + \\
&\quad T(hot, slow, hot)[R(hot, slow, hot) + \gamma V_0(hot)]
\end{aligned}
$$

$$
\begin{aligned}
Q(ok, fast) &\leftarrow 0.5[+2 + 0.8 \times 0] + 0.5[+2 + 0.8 \times 0] = 2 \\
Q(ok, slow) &\leftarrow 1[+1 + 0.8 \times 0] + 0 = 1 \\
Q(hot, fast) &\leftarrow 0.5[-10 + 0.8 \times 0] + 0.5[+2 + 0.8 \times 0] = -4 \\
Q(hot, slow) &\leftarrow 1[+1 + 0.8 \times 0] + 0 = 1
\end{aligned}
$$

Iteration 2:

$$
V_2(\text{ok}) \leftarrow \max\{Q(ok, fast), Q(ok, slow)\} = \max\{3.2, 2.6\} = 3.2
$$

$$
\begin{aligned}
Q(ok, fast) &\leftarrow T(ok, fast, ok)[R(ok, fast, ok) + \gamma V_1(ok)] + \\
&\quad T(ok, fast, hot)[R(ok, fast, hot) + \gamma V_1(hot)] \\
Q(ok, slow) &\leftarrow T(ok, slow, ok)[R(ok, slow, ok) + \gamma V_1(ok)] + \\
&\quad T(ok, slow, hot)[R(ok, slow, hot) + \gamma V_1(hot)]
\end{aligned}
$$

$$
\begin{aligned}
Q(ok, fast) &\leftarrow 0.5[+2 + 0.8 \times 2] + 0.5[+2 + 0.8 \times 1] = 3.2 \\
Q(ok, slow) &\leftarrow 1[+1 + 0.8 \times 2] + 0 = 2.6
\end{aligned}
$$

(b) Run Q-learning with a discount of .8 and a learning rate of 0.5, using the transition samples below. Do not copy over Q-values which have not changed in a given step. Assume the agent experiences the samples:
ok, fast, hot, reward +2, calculate Q_1
hot, fast, ok, reward -10, calculate Q_2
ok, slow, ok, reward +1, calculate Q_3

s	a	Q_0	Q_1	Q_2	Q_3
ok	**slow**	0			0.9
ok	**fast**	0	1		
hot	**slow**	0			
hot	**fast**	0		-4.6	

The Q-learning equation is:

$$
Q(s_t, a_t) \leftarrow (1 - \alpha) \cdot Q(s_t, a_t) + \alpha \cdot \left(r_t + \gamma \cdot \max_a Q(s_{t+1}, a)\right)
$$

SOLUTION:
$\langle ok, fast, hot \rangle$: $Q_1(ok, fast) \leftarrow 0.5 \times 0 + 0.5(2 + 0.8 \times 0) = 1$
$\langle hot, fast, ok \rangle$: $Q_2(hot, fast) \leftarrow 0.5 \times 0 + 0.5(-10 + 0.8 \times \max\{0, 1\}) = -4.6$
$\langle ok, slow, ok \rangle$: $Q_3(ok, slow) \leftarrow 0.5 \times 0 + 0.5(1 + 0.8 \times \max\{0, 1\}) = 0.9$

4. **[MDP]** Suppose you are given a simple 3-state MDP, specified with the following transition function T and the reward function R. In the table provided below, run the value iteration algorithm until the policy converges. Use the discount factor $\gamma = 0.5$. When choosing the maximum valued action for each state, break ties by choosing left before right.

Reward Function

s	$R(s)$
a	2
b	4
c	-10

Transition Model

s_{t-1}	a_{t-1}	$P(s_t = a \mid s_{t-1}, a_{t-1})$	$P(s_t = b \mid s_{t-1}, a_{t-1})$	$P(s_t = c \mid s_{t-1}, a_{t-1})$
a	left	0	0.2	0.8
a	right	0	0.8	0.2
b	left	0.8	0	0.2
b	right	0.2	0	0.8
c	left	0.2	0.8	0
c	right	0.8	0.2	0

SOLUTION:

Value Iteration

Iteration	$V(a)$	$V(b)$	$V(c)$	$\pi(a)$	$\pi(b)$	$\pi(c)$
1	0	0	0	left	left	left
2	1.2	-0.4	3.6	right	left	left
3	1.4	0.44	3.56	right	left	left
4						

Iteration 2		**Iteration 3**	**Formulas**
qa left	-7.2	-5.8	0.2*(4 + 0.5*V(b)) + 0.8*(-10+0.5*V(c))
qa right	1.2	1.4	0.8*(4 + 0.5*V(b)) + 0.2*(-10+0.5*V(c))
qb left	-0.4	0.44	0.8*(2 + 0.5*V(a)) + 0.2*(-10+0.5*V(c))
qb right	-7.6	-6.04	0.2*(2 + 0.5*V(a)) + 0.8*(-10+0.5*V(c))
qc left	3.6	3.56	0.2*(2 + 0.5*V(a)) + 0.8*(4+0.5*V(b))
qc right	2.4	2.84	0.8*(2 + 0.5*V(a)) + 0.2*(4+0.5*V(b))

For each state (a, b, c) we take the maximum of the Q-value of left and right.

There is also a python program `mdp-Q3.py` that computes 10 iterations.